

Results: Nineteen pts (95%) were treated with steroids prior to RT. Of these, 13 had a PR and 2 had a CR to pre-RT steroids. The most common pre-RT symptoms were periorbital edema (70%), visual acuity loss (60%), and pain (50%). Ten pts (50%) were on steroids at the initiation of RT. At 4 months post-RT, 17 pts (85%) demonstrated response to RT, with 7 (35%), 1 (5%), and 9 (45%) achieving a PR, CR I, and CR II, respectively. Five of 10 pts (50%) who were on steroids at initiation of RT were completely tapered off by 4 months post-RT. Pts with a response to pre-RT steroids were significantly more likely to achieve CR II with RT than those without a response ($p = 0.03$). Of the 17 pts who had $\geq$ PR at 4 months post-RT, 6 (35%) experienced recurrence of symptoms, with a median time to recurrence of 8 months. One (17%) and 3 (50%) of the recurring pts achieved a PR and CR I, respectively, with salvage therapy consisting of steroids, immunosuppressants, surgery, or a combination. Pts who achieved CR I or II by 4 months post-RT were significantly less likely to recur than those who did not ($p = 0.01$). Long-term complications were seen in 7 pts (35%), including 4 with cataract formation, 1 with chronic dry eye, 1 with enophthalmos, and 1 with keratopathy.

Conclusion: RT is an effective treatment for improving symptoms and tapering steroids in patients with biopsy-proven OP. Patients with a pre-RT response to steroids experienced significantly better outcomes at 4 months post-RT than those without a response. Additionally, complete response to RT was associated with a significantly reduced risk of symptom recurrence as compared with lesser response. The observed late complications may be related to RT, chronic use of steroids/immunosuppressants, or both.

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Treating Medium-sized Choroidal Melanomas With Eye Plaque Brachytherapy (125-I)

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Purpose/Objective(s): The purpose of this study is to compare outcomes using plaques to treat medium-sized choroidal melanomas with the published outcomes of similar tumors treated with the COMS plaques. We hypothesize that the plaques would have similar rates of local control and fewer side effects than the COMS plaques.

Materials/Methods: We conducted an IRB approved retrospective review of medium-sized choroidal melanoma patients treated from 2004-2009. Medium sized choroidal melanomas have a basal diameter of 5-16mm and apical height of 2.5-10mm. We analyzed local control, distant metastasis rate, vision preservation, and late radiation side effects to compare to the published COMS data. All of the surgeries were performed by a single ocular surgeon (Murphree) without scleral diathermy. Dr. Astrahan designed his own plaques (125I) and image based 3D treatment planning software. Patients were followed every 3-6months for the first year then every 6 months.

Results: Sixty patients were treated from 2004-2009. All patients were White. They had medium-sized choroidal melanomas and received 85 Gy to tumor apex. Median follow-up was 36.8 months (3.7years). Ten percent of patients developed metastasis. Of the 60 patients, 70% patients had 2 year and 45% had 3 yr follow-up. Local failure at three years was observed in 3% of eyes compared with 10.3% in the same time frame following treatment with COMS plaques. The following were the late radiation side effects: retinopathy (18/60), cataracts (20/60), and optic neuropathy (10/60). In terms of vision in affected eye, vision was preserved in 28/43 (65%) of patients at the 2-year follow-up. Seven percent actually had vision improvement. Only 28% had worsened vision at 2 years.

Conclusion: Use of the plaques was associated with fewer local recurrences than reported with the COMS plaques. The plaque planning

software and design of the plaques produced clinical results, which compared favorably to the COMS data. There was an equivalent distant metastasis rate of 10% in both series suggesting that the likelihood of metastatic spread is an inherent feature of the melanoma and not related to treatment technique. Radiation related late side effects are similar to the COMS data. However, preliminary results suggested that vision preservation was better with the eye plaques (28% had worsened vision in our study as compared with 49% of COMS patients). Future studies and larger numbers of patients will be required to confirm the low tumor recurrence rates and the encouraging visual outcomes.

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Stereotactic Radiosurgery in the Treatment of Ocular Melanoma

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Purpose: To report our experience and evaluate our results using stereotactic radiosurgery to treat uveal melanoma in patients who were not suitable for standard treatment with iodine-125 plaque brachytherapy or enucleation.

Materials and Methods: Between 2000 and 2010, 32 patients were treated using stereotactic radiosurgery for uveal melanomas. Six patients were treated with a frame-based unit, and 26 patients were treated with a frameless system. Twenty-nine patients were treated in a single fraction with radiation doses ranging from 14-20 Gy, and three patients were treated with fractionated regimens (30 Gy in 5 fractions and 48 Gy in 8 fractions) for a larger lesion. Patient charts were reviewed for clinical and treatment characteristics.

Results: Median follow-up was 13.5 months (range, 2- 42 months). Twenty-three patients presented with symptoms, with a median duration of 2 months (range, 0-15 months). At diagnosis, the mean tumor thickness was 6.36 mm (range, 1.5-14.4 mm). Mean thickness after treatment was 5.25 mm (range, 1-13.5 mm), a reduction of 17%. The 24-month overall survival was 65%, and cancer-specific survival was 88%. Local control was 94%, and distant control was 91%. Three patients (9%) reported eye pain following treatment, two severe enough to require enucleation. Seven patients (22%) had vitreous hemorrhage, and nine (28%) developed radiation-induced cataracts. Nine patients (28%) developed radiation retinopathy and six (19%) developed optic papillopathy. Four patients (13%) developed iris neovascularization, two patients (6%) developed neovascular glaucoma, and one patient (3%) developed radiation-induced ocular ischemia. No patients developed sclera necrosis.

Conclusion: Stereotactic radiosurgery is a safe, well-tolerated, and effective treatment technique for patients with uveal melanomas that are not amenable to standard treatment with brachytherapy plaque treatment or enucleation.

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Predictors of Tumor Recurrence and Enucleation After Iodine-125 Plaque Brachytherapy for Iris Melanomas

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Purpose: To identify predictive variables of outcome for patients treated with Iodine-125 plaque radiation therapy for iris melanomas.